

Figure 3.22 Creating a field in an index with a custom BM25 similarity function

We create a mapping definition, assigning our custom similarity function (`custom_BM25`) to a `synopsis` field. When ranking results based on this field, Elasticsearch considers the provided custom similarity function to apply the scores.

> **Similarity algorithms are beasts**
>
> Information retrieval is a vast and complex subject. The algorithms involved in scoring and ranking results are advanced and complicated. While Elasticsearch provides a few similarity algorithms as plug-and-play, you may need a deeper understanding when working with them. Once you have tweaked the configurational parameters of these scoring functions, be sure you have tested and tried every possible combination.

You may wonder how Elasticsearch can retrieve documents in a fraction of a second. How does it know where in these multiple shards the document exists? The key is the routing algorithm, discussed next.

3.5 Routing algorithm

Every document has a permanent home: that is, it must belong to a particular primary shard. Elasticsearch uses a *routing algorithm* to distribute the document to the underlying shard when indexing.

Routing is a process of allocating a home for a document to a certain shard, with each document stored in one and only one primary shard. Retrieving the same document is easy, because the same routing function is used to find the shard to which that document belongs.

The routing algorithm is a simple formula that Elasticsearch uses to deduce the shard for a document during indexing or searching:

```
shard_number = hash(id) % number_of_shards
```

The output of the routing function is a shard number. It is calculated by hashing the document's ID and finding the remainder of the hash when divided (using the modulo operator) by the number of shards. The `hash` function expects a unique ID, generally a document ID or a custom ID provided by the user. Documents are evenly distributed, so there is no chance of one of the shards being overloaded.